

SEUNGMAN CHOI

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EDUCATION & WORKING

Purdue University

PhD. in electrical and computational engineering (GPA 3.94/4.00)

Advisor: [Zubin Jacob](#)

Super-resolution imaging through adaptive information mining [6,9–12,14].

Computational imaging detection and ranging (CIDAR) [15].

West Lafayette, IN

Aug 2022-May 2027 (expected)

Korea Advanced Institute of Science and Technology (KAIST)

Research associate in Mechanical engineering

Advisor: [Seung-Woo Kim](#), [Young-Jin Kim](#)

EUV tool development for semiconductor inspection [3–5,8].

Daejeon, KR

Apr 2019- Aug 2022

Tokyo Institute of Technology (Ti-tech)

M.S & B.S in Mechanical Engineering (Magna Cum Laude)

Advisor: [Nisisako Takasi](#), [Shigeki Saito](#)

Scaled-up Microfluidic device using semiconductors manufacturing technique [7,13].

Advanced gripping system for semiconductors with complex shapes [1,2].

Tokyo, JP

Apr 2013- Mar 2019

RELATED PUBLICATIONS LIST

‡: Equally contributed to works.

[15] S. Jape, [S. Choi](#), et al., “Night vision transformers: hyperspectral reconstruction and passive depth estimation from a single broadband thermal image”, In preparation.

[14] [S. Choi](#), et al., “Decoding atmospheric turbulence for few-frame diffraction-limited imaging”, In preparation.

[13] T. Nisisako, [S. Choi](#), et al., “Mechanically and directionally tunable step emulsification via in-plane stretching of an elastomeric microfluidic device”, Under review at *Ind. Eng. Chem. Res.*

[12] [S. Choi](#), et al. “Adaptive information mining: Approaching the quantum limit of super-resolution imaging”, Under review at *Optica*.

[11] P. Menart†, [S. Choi†](#), et al., “Definition of multi-parameter super-resolution for optical telescopes”, Under review at *Optics Express*.

[10] [S. Choi](#), et al., “Poisson wavefront imaging in photon-starved scenarios”, *J. Opt. Soc. Am. B*, 43, 231-238 (2026).

[9] [S. Choi](#), et al., “Photon-starved imaging through turbulence at the diffraction limit”, <https://arxiv.org/abs/2510.22806>, 2025.

[8] S. Won, J. Kim, Y. Oh, T. Kim, [S. Choi](#), et al., “Programmable structured DUV illumination by coherent harmonic generation at crystalline solids for nanometer-resolution inspection of periodic samples” *Photonix*, 6(1), 1-15, 2025.

[7] N. Tottori, [S. Choi](#), et al., “Production of monodisperse oil-in-water droplets and polymeric microspheres below 20 μm using a PDMS-based step emulsification device”. *Micromachines*, 16(2), 132, 2025.

[6] H. Choi†, [S. Choi†](#), et al., “Telescope imaging beyond the Rayleigh limit in extremely low SNR”, *New Journal of Physics* 26 (9), 093019, 2024.

[5] B. Kim†, [S. Choi†](#), et al., “Coherent manipulation of extreme-ultraviolet Bessel vortex beam from solids by active wavefront shaping of driving fundamental beam” *ACS Photonics*, 10.1021/acsphtoncis.3c00645, 2023.

[4] S. Won†, [S. Choi†](#), et al., “Dynamic ultraviolet harmonic beam pattern control by programmable spatial wavefront modulation of near-infrared fundamental beam” *Nanophotonics*, 12 16, 2023.

[3] Y. Kim†, B. Kim†, [S. Choi†](#), et al., “Compensation of laser propagation effects within solids for high harmonic generation of extreme ultraviolet radiation” *Opt. Laser Technol.*, 148 107507, 2022.

[2] Y. Taoka, K. Kawabata, P. Hemthavy, [S. Choi](#), et al., “Development of Bipolar Electrostatic Chuck with a Beam-Array Assembly Fabricated by Lithography”. *International Journal of Automation Technology*, 16(4), 471-477, 2022.

[1] [S. Choi](#), et al., “Development of a bipolar electrostatic chuck module with a beam-array assembly using the multiple etching process”, *J. Micromech. Microeng.* 28 125011, 2018,

HONORS, AWARDS & GRANTS

[13] **Bilsland Fellowship** Jan 2027–Mar 2027
Purdue University, \$31,000 for a semester, US.
[12] **DARPA CIDAR Semifinal Winner** Jan 2026
Zetascope, [\\$50,000 team award](#), US. Core member;
[closest final-round result](#) to DARPA's winning criteria.
[11] **Outstanding Presentation Award** Feb 2025
The 10th 2025NCC, KR.
[10] **Outstanding Presentation Award** Nov 2024
Autumn 2025KSPE, KR.
[9] **Best Presentation Award** May 2023
The 18th 2023IEEE-NEMS, KR.
[8] **Best Paper Award** Nov 2022
The 9th 2022ASPEN, SG.
[7] **Bronze Prize** Nov 2022
Autumn 2022KSME, KR.

[6] **Best Paper Award** Nov 2021
Next generation lithography conference (2021OSK), KR.
[5] **Excellence Presentation Award** Feb 2019
2019KSEAJ, JP.
[4] **ITO scholarship** Apr 2017–Mar 2019
ITO Foundation for International Education Exchange,
JPY 180,000 per month, JP.
[3] **Tour grant** Aug 2018
The Precise Measurement Technology Promotion
Foundation (PMTF-F), JPY 197,000 once, JP.
[2] **MutoEiji academic excellence award** Mar 2017
The Japan Society for Design Engineering (JSDE), JP.
[1] **KR-JP joint scholarship** Apr 2013–Mar 2017
Korea and Japan Government, JPY 130,000 per month,
KR&JP.

PREVIOUS RESEARCH INTERESTS

Keywords: Computational adaptive optics, information-optimal sensing, aperture-synthesis, computational imaging, physics-informed learning

Information-optimal computational imaging Design

Optical measurements to maximize useful information under photon, noise, and hardware constraints.

- Optimal sensing: Co-design optical encoding and aperture for detection, localization, and parameter estimation.
- Adaptive measurements: Reconfigure sensing according to the scene, aberration state, or inference objective.
- Physics-informed decoding: Combine wave-optical models with compact learned representations for efficient reconstruction.

Computational adaptive optics

Jointly recover high-resolution scenes and time-varying optical aberrations from coded measurements.

- Aberration correction: Estimate atmospheric turbulence and system aberrations directly from intensity measurements.
- Spatiotemporal inference: Exploit spatial and temporal structure to recover rapidly evolving wavefronts
- Compact aberration models: Represent optical dynamics using physical bases and lightweight models such as Zernike modes and MLPs.

SKILLS

Programming language	Python, Matlab and LabView
Optical system design	Vacuum, microscopic and astronomical telescopic optical system
Computational algorithm	Coherent diffraction imaging (Ptychography), double blind-deconvolution, deep image prior and genetic algorithm.
Clean room nanofabrication	Deep reactive-ion etching and photo-lithography